

SOFTWARE ENGINEERING

LAB 3: Architecture Justification

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SYSTEM: Telemedicine Slot Booking & Prescription Portal (Problem Statement #11)

Architecture Selection

The telemedicine slot booking and prescription portal system will adopt a layered architecture. The system activities will be structured into three layers namely: Presentation (user interface component), Business (booking manager, payment service, prescription service, and video session), and Data (database component) with down-stream dependencies as shown below. This approach supports the key system workflows as described below:

Reason 1: The booking workflow is a tight transaction

The process identified in Lab 1 and elaborated in Lab 2 is a single sequential transaction of activities starting with browsing specialties and ending with the encrypted link for video consultation. Each activity is a prerequisite to the next and does not yield value to the user until the end of the transaction. The layered approach supports this requirement since the booking manager component can perform the transaction within the business layer without breaking it into multiple distributed transactions that would require coordination and compensating mechanisms to maintain consistency between independent services.

Reason 2: Centralized booking state management prevents double-booking

As required by FR-003, the system should not allow a slot to be booked by two patients. The layered approach supports this requirement since the booking state (i.e., temporarily held or confirmed) is managed within the business layer. Consequently, all booking-related activities (temporary slot hold, confirmation) are executed against a single database component through the IDataAccess interface. Under a distributed architecture, the same level of consistency would require complex compensating transactions to ensure that the booking state is visible across independent services.

Security benefit: Reduced number of trust boundaries

The layered approach will be designed with a small number of trust boundaries, thus supporting the security requirements NFR-001 (TLS 1.3 for data in transit and AES-256 for data at rest). Specifically, the user interface component (presentation layer) will have a single trust boundary with the booking manager through the IBookingRequest interface. The business layer will have another trust boundary with the database component through the IDataAccess interface in support of NFR-001. Overall, this approach reduces the system's attack surfaces while simplifying the encryption, authentication, and auditing requirements for protected health information (PHI). Notably, the video session, prescription service, and payment service components will be subject to encryption and user authentication checks at two trust boundaries.

Performance benefit: Fewer network hops

Components within the same process boundary do not require network communication. Thus, the layered approach will help minimize unnecessary network hops between components, thereby improving performance. As discussed earlier, the slot booking process (FR-003) requires a temporary slot hold for no more than 10 minutes. Similarly, the system should issue the encrypted link (FR-004) no later than 10 minutes to the scheduled consultation time. This is critical since it gives the patient enough time to complete the payment process (payment service) and obtain the encryption key (prescription service) before accessing the video consultation link. However, the payment process requires a network hop and cannot be handled within the business layer due to the involvement of an external payment gateway.

Components

- **User Interface Component:** The presentation layer is responsible for patient and physician interaction and is mapped to FR-002 and FR-003.
- **Booking Manager Component:** This component is responsible for handling slot selection, temporary slot hold, and confirmation and is mapped to FR-003.
- **Payment Service Component:** This component interfaces with an external payment gateway to process payments for video consultation.
- **Prescription Service Component:** This component is responsible for digitally signing video consultation notes and generating prescription PDFs and is mapped to FR-001 and FR-005.
- **Video Session Component:** This component is responsible for generating encrypted links for video consultation and is mapped to FR-004.
- **Database Component:** The data layer is mapped to NFR-001 (encryption at rest using AES-256) and stores slot booking information, consultation notes, and prescription information.

Trade-off

The video session component cannot be independently scaled-out or load-balanced. If the telemedicine system experiences significant growth and scalability challenges emerge, the video session component should be extracted and exposed as an independent service API. At that point, the system can transition to either a service-oriented or microservice architecture.